

ECON 326 Final Project

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Apr. 9, 2025

abstract

Abstract This project explores how different aspects of students' daily routines are connected to the hours of sleep they get. While the answer might seem obvious at first, uncovering the actual relationships requires a thoughtful approach using regression analysis. The data was collected from students in UBC's ECON326 course. The analysis starts with selecting variables and filtering out those that were irrelevant or showed high multicollinearity, based on results from theoretical reasoning and correlation outputs from R. Several variables are selected as main predictors that we believe could negatively affect sleep hours. We interpret regression results based on model fitness and confidence interval estimation. After trimming data through identifying and removing outliers, we were able to find that hours spent on assignments and class preparation, as well as financial aid a student received, had a negative and statistically significant impact on sleeping hours. We then discussed possible channels through which these variables had an impact on sleep, using existing literature.

1 Introduction

Sleep duration and quality is an important factor in human health. Individuals who sleep less than six hours a night have a 20 percent higher risk of heart attacks, and these effects have been shown to compound the longer an individual goes without the recommended 7-8 hours of sleep (Daghlal et al., 2019). The quality of sleep is also important, as restless sleep has been shown to increase stroke prevalence compared to restful sleep that follows the normal patterns of REM (rapid eye movement) and deep sleep cycles (Zhou et al., 2020). This is especially crucial for students, who must be alert and wakeful to properly participate in academic and extracurricular activities. Unfortu-

nately, many students report poor sleep quality due to various lifestyle choices. This includes issues like noisy or brightly lit sleeping environments, lack of awareness about the importance of sleep, stressors, or excessive screen time. A lack of a proper sleep environment correlates with decreased sleeping hours, as more time is spent in a wakeful state at night. (Yao & Wang, 2023). In our research paper, we are interested in analyzing the effects of various routines and habits on sleep hours, as a proxy for sleep quality, reported by students using a multiple OLS regression.

2 Literature Review

There is a variety of literature on the relationship between sleep hours and variables like exercise, screen usage, study habits, and quality of social relationships. A majority of academic literature on sleep utilizes the Pittsburgh Sleep Quality Index (PSQI) to report sleep quality. Subjects self report their sleeping habits, disturbances, and general satisfaction with each night, and are scored from 0 to 21. A score of 5 or more represents issues with sleep quality, and higher scores generally imply poorer sleep quality. Time asleep is a key variable in this index. Although our research question will be focused on sleeping hours, we will review literature that utilizes this index in addition to hours asleep since the PSQI is so widely used, and it is difficult to extrapolate the effects of sleep hours on their own, as psychiatric papers require more robust and holistic approaches to measuring sleep. However, we argue that this does not pose an issue for our research question pertaining to the impact of sleep hours specifically, as the index, apart from sleep hours, also measures time in bed, number of times an individual wakes up throughout the night, which all impact sleep in a negative way by shortening its span, and therefore still correlate with sleep hours. Therefore, our use of sleep hours as a dependent variable

is a reasonable assumption based on existing research using the PSQI. (Shahid et al., 2011).

Students are especially impacted by poor sleep quality, which has been shown to decrease academic performance and increase stress rates (Ahrberg et al., 2012). Additionally, adolescents that experience decreased sleep quality suffer from increased likelihood of depression symptoms. These effects compound, and every night of poor sleep leads to increased prevalence of mental health issues (Tokur-Kesgin Kocoglu-Tanyer, 2021).

Sleep quality can either be improved or worsened by routines and habits of students. For example, students can have their sleep schedules negatively affected by study decisions. Gillen-O’Neel et al. found that students regularly sacrifice sleep hours to study more, however this decision results in outcomes that are worse than if the student had slept the additional hours (2013). Additionally, compulsive smartphone usage, whether through constant social media usage or other screen-related activities, was shown to increase technostress, or the stress associated with usage of information and communication technologies. Increased technostress led to worse sleep quality outcomes for Chinese university students (Yao Wang, 2022).

Some routines, however, have been shown to have a positive effect on sleep quality. Notably, Chen et al. demonstrated in “Association of physical activity and positive thinking with global sleep quality” that college students who regularly engaged in vigorous physical activity, such as aerobic exercise, as well as engaged in some positive thinking, have seen improvements in sleep quality (2022). Additional research extracts the effect of physical research only, finding positive effects of exercise on sleep quality, especially for first-year students (Schori, 2022). In addition to exercise, socialization may also improve sleep quality outcomes. University students who reported greater social stress had higher (worse) PSQI scores, while students with a diverse social integration, such as a greater

number of local friendships, reported lower (improved) PSQI scores (Jin et al., 2014).

In summary, a greater variety of stressors, whether academic, social, or technological, lead to worse sleep quality, and therefore decreased sleeping hours, while improved physiological well-being through exercise, as well as improved social well-being and satisfaction through increased friendships lead to increased sleep quality, and therefore sleeping hours.

3 Research Question

While university students are often used as subjects in sleep research, there is little research however, on the combination of lifestyle factors and their impacts on sleeping hours. We wish to fill this research gap by examining how multiple variables at once impact the sleeping hours of university students. Existing economic literature, such as research by Jin and Ziebarth, focused on the detrimental effects of a lack of sleep on productivity and economic outcomes (2020). However, there is little economic research on the causes of such sleep deprivation. We wish to combine existing psychiatric, economic, and medical literature into a single study to observe the strength of the coefficients on each variable, and find what variables have a greater positive or negative impact on sleeping hours. We have formulated our research question to be the following:

“How do students’ daily routines affect sleep hours?”

4 Hypothesis

- **Non-academic screen time and sleep (#significant)**
 - H: Increased time spent on social media has no effect on hours slept.

- H: Increased time spent on social media has a significant effect on hours slept.
- In this scenario, our null hypothesis is that social media has no impact on hours slept, while our alternate hypothesis is that social media use significantly impacts sleep hours.

- **Exercise and sleep**

- H: Increased exercise hours have no effect on hours slept.
- H: Increased exercise hours have a significant effect on hours slept.
- In this scenario, our null hypothesis is that exercise has no impact on hours slept, while the alternate hypothesis is that it significantly affects sleep duration.

- **Socialization and sleep**

- H: Increased hours spent socializing have no effect on hours slept.
- H: Increased hours spent socializing have a significant effect on hours slept.
- Here, the null hypothesis states that socialization does not impact hours slept, while the alternate claims it does have a significant effect.

- **Study time and sleep**

- H: Increased study hours have no effect on hours slept.
- H: Increased study hours have a significant effect on hours slept.
- In this case, we test whether study time influences sleep, assuming no effect under the null and a significant effect under the alternate.

We expect that all the main predictors would have negative coefficients, since the trade-off between sleep hours and daily activity deprive students' sleeping hours are commonly observed in reality.

5 Data and sample

5.1 Data source

For this study, we will use the data from Track 1, focusing on students' sleep hours as the dependent variable and the daily routine factors such as study hours, socialization hours, screen time, and exercise hours as independent variables. This dataset allows us to examine the relationship between students' daily routines and their sleep duration, which serves as a control for sleep quality. This data was collected through an anonymous survey of UBC undergraduate Economics students, complying with all ethical and privacy standards outlined by the University of British Columbia.

5.2 Data selection

Based on theoretical reasoning and data relevance, we selected the following variables as main predictors to represent students' daily routines: `hour_all_courses` (total hours spent on coursework), `hour_social` (combined hours spent socializing with friends and family), `hour_screen_nonacademic` (non-academic screen time), `hour_sport` (exercise hours), and `hour_work` (time spent working).

Several variables were removed due to multicollinearity due to their redundancy in theoretical explanation,. For example, GPA on the 4.0 scale was removed due to multicollinearity with GPA variable in percentage; course-specific time variables like hours on ECON326, class materials, and assignments were consolidated into `hour_all_courses`; and variables such as transportation mode,

scholarship details, and feelings about AI, future employment, and financial situation were excluded due to lack of direct relevance to our analysis. Notably, we combined time spent socializing with friends and family into a single variable as `social_hours` based on the previous research that there is minimal overlap between the two and that they collectively capture social interaction.

After selecting the most relevant regressors based on theoretical reasoning, we examined potential multicollinearity among the predictors. To do this, we generated a GGpairs plot in R, and the results are shown in Table 1. From the plot, we observed a high correlation between social media hours and non-academic screen time hours. Since non-academic screen time hours should also include the hours spent on social media, social media hours was dropped from our model. This can be seen in pper-Triangle Correlation Matrix below.

Table 1: Upper-Triangle Correlation Matrix

Variable	GPA	Hour_all_courses	Extra_hour_forecam	Coursestake_semester	AL_tool_Hour	hour_sports	hour_studentsclub	hour_volunteer	hour_socialize	hDomestic	percent_financial_aid	Hour_work	sleep_hours	hour_social_media	Hour_screen_timeacademic	feeling_depress	Hour_commute
GPA	-																
Hour_all_courses	0.048	-															
Extra_hour_forecam	0.048	0.048	-														
Coursestake_semester	0.048	0.048	0.048	-													
AL_tool_Hour	0.048	0.048	0.048	0.048	-												
hour_sports	0.048	0.048	0.048	0.048	0.048	-											
hour_studentsclub	0.048	0.048	0.048	0.048	0.048	0.048	-										
hour_volunteer	0.048	0.048	0.048	0.048	0.048	0.048	0.048	-									
hour_socialize	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	-								
Domestic	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	-							
percent_financial_aid	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	-						
Hour_work	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	-					
sleep_hours	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	-				
hour_social_media	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	-			
Hour_screen_timeacademic	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	-		
feeling_depress	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	-	
Hour_commute	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	0.048	-

Variable Name	Description
GPA	The student's cumulative grade average, reported as a percentage. This serves as the dependent variable in the regression model.
Hour_all_courses	The estimated number of hours per week a student spends studying across all their courses before the exam period.
Extra_hour_forexam	Additional hours spent studying in preparation for exams, beyond regular weekly coursework.
Coursetake.semester	The number of academic courses the student is enrolled in during a given semester.
AL_tool_Hour	Hours spent each week working on assignments, exercises, and other academic tasks with AI tools.
hour_sports	Weekly hours students dedicate to physical activities such as sports or fitness.
hour_studentclub	Hours per week that students participate in student clubs or societies.
hour_volunteer	Time spent each week on volunteering or community service.
hour_socialize	Total weekly hours spent socializing with friends and/or family, either in-person or online.
IsDomestic	A dummy variable indicating if a student is domestic (1) or not (0). The original categorical responses were recoded for use in regression.
percent_financial_aid	The percentage of tuition covered by financial aid or scholarships. If the student receives no aid, this value is 0.
Hour_work	Hours the student works per week while studying. A value of 0 indicates the student does not work.
sleep_hours	The average number of hours of sleep the student gets per night.
hour_social_media	Time (converted into hours) that students spend daily on social media platforms.
Hour_screen_nonacademic	Daily hours spent using screens (phones, TVs, computers) for non-academic purposes.
feeling_depress	A self-reported scale from 1 to 5 measuring how frequently the student feels depressed or anxious.
Hour_commute	Time spent commuting to UBC (in hours). This was converted from minutes in the raw data. The conversion does not affect the R^2 or the variance of the variable.
year_level	The student's year level in their undergraduate program (e.g., 1st year, 2nd year, etc.).

Table 2: Descriptions of Variables Used in the Regression Model

(1) According to previous research conducted by Christopher RENNICK from the University of Waterloo in 2020, researchers tended to separate the time spent with family and friends into different categories. Based on this precedent, we believe it is reasonable to combine both and treat the total time spent with friends and family as one variable—*hour_socialize*. (2) The following variables were dropped from the model due to issue of multicollinearity or irrelevance: GPA on 4.0 scale (multicollinearity with GPA), preference for practical or theoretical concepts (irrelevant), hours spent on ECON326, primary mode of transportation (we think that hours in transportation matters more), feel of AI (irrelevant), Hour.class.material and Hour.assignment (both are part of Hour_all_courses), tuition (could be similar to IsDomestic), scholarship (same as percentage financial aid), faculty, work status and job type (we use Hour_work instead), feeling about financial situation, course delivery preference, concerns about future employment, future education (might be multicollinear with GPA), career path, and ethnicity (where do students come from). (3) We only used one of social media hours and nonacademic screentime hours to avoid the multicollinearity

6 Baseline regression model

To begin with, we ran the baseline regression model including all relevant predictors. These factors were selected based on theoretical reasoning, including students' exercise hours (*hour_sports*), hours spent with family and friends (*hour_socialize*), total hours spent on all courses (*Hour_all_courses*), and non-academic screen time (*Hour_screen_nonacademic*). Our variable of interest is how students' daily time allocation affects their sleep duration.

$$\begin{aligned} \text{sleep_hours} = & \beta_0 + \beta_1 \cdot \text{hour_sports} + \beta_2 \cdot \\ & \text{hour_socialize} \\ & + \beta_3 \cdot \text{Hour_all_courses} + \beta_4 \cdot \\ & \text{Hour_screen_nonacademic} + \epsilon \end{aligned} \quad (1)$$

Variable	Est.	SE	t	p
(Intercept)	7.3476	0.3000	24.49	<2e-16 ***
hour_sports	-0.0215	0.0254	-0.85	0.3985
hour_socialize	-0.0015	0.0124	-0.12	0.9058
Hour_all_courses	-0.0002	0.0103	-0.02	0.9817
Hour_screen_nonacademic	-0.0433	0.0247	-1.76	0.0811

SE = 1.255, DF = 163
 $R^2 = 0.024$, Adj. $R^2 = 0.0003$
 F = 1.013, p = 0.4025

Table 3: Sleep Model (Compact)

After running the full regression, we found no major statistically significant relationships between the main predictors of daily routine and sleep hours. The R^2 is less than 0.05, suggesting the low relationship between the main predictors and Y. The lowest adjusted R^2 suggest there is issue on overfitting, which means we may added variable that is irrelevant.

Also, as F-test result is low and the probability of F-test is greater than 0.05, suggesting that the main predictors have low joint probability on Y. The p-values for each individual coefficient were relatively high, indicating insignificance.

7 Full model with controls

Then, we ran the full model with all controls that may affect sleep hours in the table 3.

$$\text{sleep_hours} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon \quad (2)$$

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.3122	1.1838	5.332	3.48e-07 ***
GPA	0.0105	0.0094	1.113	0.2674
Hour_all_courses	0.0041	0.0118	0.346	0.7299
Extra_hour_forexam	0.0028	0.0125	0.220	0.8264
Coursetake.semester	0.0092	0.1736	0.053	0.9576
AL.tool.Hour	-0.0329	0.0221	-1.493	0.1375
hour_sports	-0.0166	0.0266	-0.622	0.5349
hour_studentclub	0.0071	0.0279	0.254	0.7996
hour_volunteer	0.0071	0.0509	0.139	0.8893
hour_socialize	0.0019	0.0136	0.142	0.8876
IsDomestic	-0.1183	0.2257	-0.524	0.6008
percent_financial_aid	0.0038	0.0067	0.565	0.5729
Hour_work	-0.0150	0.0175	-0.853	0.3948
Hour_screen_nonacademic	-0.0440	0.0257	-1.714	0.0886 .
feeling_depress	-0.0257	0.0939	-0.273	0.7851
Hour_commute	0.0141	0.0087	1.614	0.1086
year_level	0.0412	0.1390	0.296	0.7673

Residual SE: 1.27 on 151 DF
 $R^2 = 0.0742$, Adjusted $R^2 = -0.0239$
 F-statistic: 0.756 on 16 and 151 DF, p-value: 0.7324

Table 4: Full Regression Model: Predicting Sleep Hours from All Available Variables

The adjusted R^2 is low, suggesting a poor fit for the model. However, in the full model—which includes all reasonable predictors—the standard error of the coefficients for the main predictors increased by only 0.1% compared to the baseline model. This minimal change suggests that there is low multicollinearity among the variables. Since the coefficients are not sensitive to the inclusion of additional controls, it supports the conclusion that multicollinearity is not a major concern in the regression with the main predictors.

7.1 Full model - F-statistic

The F-statistic with degrees of freedom $F_{(16, 151)}$ approaches the critical value of 1.80, but the actual F-statistic from the model is much lower, resulting in an insignificant p-value of 0.7324—well above the typical significance level of 0.05. Thus, the independent variables taken together (including the main predictors and control variables) do not explain a significant portion of the variation in the dependent variable.

Table 5: ANOVA Comparison of Baseline and Full Regression Models

Model Formulas

Model 1: `sleep_hours ~ hour_sports + hour_socialize + Hour.all.courses + Hour.screen.nonacademic`
 Model 2: `sleep_hours ~ GPA + Hour.all.courses + Extra.hour.forexam + Coursetake.semester + AL.tool.Hour + hour_sports + hour.studentclub + hour.volunteer + hour_socialize + IsDomestic + percent.financial.aid + Hour.work + Hour.screen.nonacademic + feeling.depress + Hour.commute + year.level`

Res.Df	RSS	Df	Sum Sq	F	Pr(> F)
163	256.54				
151	243.42	12	13.124	0.6784	0.7702

Then, we use the ANOVA test, which checks whether adding new variables to a simple regression improves the model's fit. The null hypothesis is that all the additional predictors in Model 2 have coefficients equal to zero. In other words, they do not contribute significantly to explaining the variation in sleep hours. The p-value of 0.7702 suggests that, assuming the added variables do not improve the model, there is a 77% probability of observing a drop of 13.124 in the sum of squares ($Sq = 13.124$), which is greater than the significance level of 5% ($p = 0.05$). Thus, we conclude that adding those variables does not lead to a significant improvement in explaining the variation in sleep hours. In the presence of all data points, the control variables do not contribute meaningfully to the model's explanatory power.

7.2 Heteroskedasticity test

We conducted a Breusch-Pagan test to check for heteroskedasticity in the full regression model. The test was implemented using the following R code:

```
# Run Breusch-Pagan tests
bp_full <- bptest(sleep_hour_model_full, ~ fitted(sleep_hour_model_full) +
                  I(fitted(sleep_hour_model_full)^2))
bp_main <- bptest(sleep_hour_model1, ~ fitted(sleep_hour_model1) +
```

```
I(fitted(sleep_hour_model1)^2))
```

Listing 1: Breusch-Pagan Test on Full and Main Predictor Models

We use the BP test to check whether the full model and the model with main explanatory variables violate the assumption that there is no heteroskedasticity in the data. We simplified that model based on the theorem that the Breusch-Pagan test is used to check if the variance of the error is correlated with some variables (otherwise it will be a constant, which is homoskedastic). Running the full model still allows us to check if the residual variance is related to the square of predicted values.

The result shows:

Table 6: Breusch-Pagan Test Results

Model	BP Statistic	df	p-value
Full Model	1.0121	2	0.6029
Main Predictors Only	4.6260	2	0.0999

We found that in the full model, the p-value of the F-test statistically significant at greater than 0.05, indicating the issue of heteroskedasticity in the full model. Thus the previous inference on hypothesis and confidence intervals may be unreliable as the model may lead to biased standard errors. It also suggests that we need to heteroskedasticity-robust standard errors later on.

The p-value of regressions with main predictors is above 0.05, so the issue of heteroskedasticity is not present at the 5% significance level (which is the threshold we primarily use). This result is consistent with the Residuals vs. Fitted Values plot generated from the R code. In other words, at the 5% significance level, we fail to reject the null hypothesis of heteroskedasticity as there is no strong evidence of it. However, there is some mild evidence of heteroskedasticity at the 10% significance level. Thus both of the test

result from BP test indicate the need to introduce heteroskedasticity-robust standard errors.

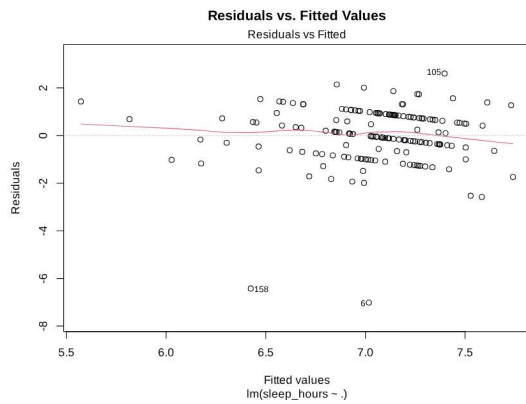


Figure 1: Residuals vs. Fitted Values plot

The Residuals vs. Fitted Values plot to visually inspect the presence of heteroskedasticity. From the graph, we observe that the residuals appear randomly scattered with no clear pattern. As the red line is nearly flat, that the variance of the residuals remains roughly constant across fitted values.

Thus we conclude that the issue of heteroskedasticity exists but mildly in the full-model regression.

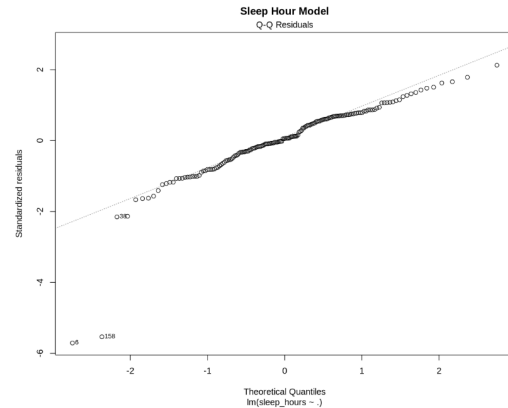


Figure 2: Quantile-Quantile Table

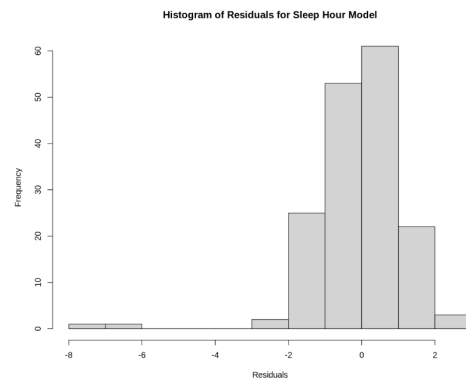


Figure 3: Histogram

The QQ-plot and histogram of residuals shows that the residuals deviate from normality at both tails, suggesting the outliers may be present at the upper and lower tail.

So if we move these outliers, our residuals appear much closer to a normal distribution, and the data would more closely match the conditions of the CLT.

8 Normality and outliers

Thus we are going to adjust our statistical method - to be more specific, we are going to use the Heteroskedastic - Robust model. But before that, we use the QQ plot to check for normality

9 Heteroskedasticity-Robust Regression (for robustness)

We have used coeftest test to recalculate the coefficient of full model regression with hetero-robust standards.

Table 7: Robust Standard Errors: Full Model Regression Results

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.3122	1.1079	5.6977	<0.00001 ***
GPA	0.0104	0.0089	1.1806	0.2396
Hour_all.courses	0.0041	0.0093	0.4365	0.6631
Extra_hour_forexam	0.0028	0.0078	0.3531	0.7245
Coursetake.semester	0.0092	0.1646	0.0561	0.9553
AL_tool_Hour	-0.0393	0.0192	-1.7133	0.0887
hour_sports	-0.0166	0.0479	-0.3459	0.7299
hour_studentclub	0.0071	0.0200	0.3726	0.7100
hour_volunteer	0.0071	0.0433	0.1638	0.8701
hour_socialize	0.0019	0.0173	0.1117	0.9112
IsDomestic	-0.1183	0.2385	-0.4960	0.6206
percent.financial.aid	0.0037	0.0092	0.4106	0.6820
Hour_work	-0.0150	0.0134	-1.1176	0.2655
Hour_screen.nonacademic	-0.0440	0.0211	-2.0633	0.0408 *
feeling_depress	-0.0257	0.0915	-0.2804	0.7796
Hour_commute	0.0141	0.0076	1.8611	0.0647
year_level	0.0412	0.1688	0.2440	0.8075

Note: * $p < 0.05$, . $p < 0.1$

Using heteroskedasticity-robust standard errors, the variable `Hour_screen_nonacademic` was statistically significant at 10%. This indicates that, holding other factors constant, students who spend more time on non-academic screen activities tend to sleep less. Specifically, each additional hour of non-academic screen use is associated with an estimated 0.044-hour (approximately 2.6 minutes) decrease in sleep. Medical research points to a negative effect of screen time on sleep, especially when adolescents use screens such as computers, phones, and tablets in the evening or nighttime, prior to going to bed (Hartley, 2022).

10 Another way of Robustness check: Trimmed data

As mentioned in the previous analysis, the QQ-plot and histogram of residuals show that the residuals deviate from normality at both tails, suggesting that outliers may be present at the upper and lower extremes.

To address this, we removed these potential outliers, and the residuals appeared much closer to a normal distribution. This adjustment brings the data more in line with the assumptions of the Central Limit Theorem (CLT).

Table 8: Regression Results Using Trimmed Data

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.1992	0.8279	8.696	< 2e-14 ***
GPA	0.0083	0.0065	1.281	0.2023
Hour_all.courses	0.0029	0.0084	0.349	0.7275
Extra_hour_forexam	0.0038	0.0087	0.432	0.6667
Coursetake.semester	-0.0171	0.1231	-0.139	0.8900
AL_tool_Hour	-0.0322	0.0154	-2.088	0.0386 *
hour_sports	0.0298	0.0204	1.460	0.1466
hour_studentclub	-0.0124	0.0194	-0.639	0.5240
hour_volunteer	-0.0119	0.0357	-0.333	0.7397
hour_socialize	-0.0091	0.0097	-0.945	0.3462
IsDomestic	0.0145	0.1629	0.089	0.9292
percent.financial.aid	-0.0138	0.0067	-2.075	0.0398 *
Hour_work	-0.0197	0.0122	-1.618	0.1078
Hour_screen_nonacademic	-0.0284	0.0179	-1.582	0.1158
feeling_depress	-0.0705	0.0683	-1.031	0.3043
Hour_commute	0.0067	0.0063	1.063	0.2897
year_level	-0.1227	0.0993	-1.235	0.2188

Residual standard error: 0.8737 on 140 degrees of freedom

$R^2 = 0.1396$, Adjusted $R^2 = 0.0413$

F-statistic: 1.42 on 16 and 140 DF, p-value: 0.1404

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

After trimming the top and bottom 5% of observations in `sleep_hours` to exclude extreme values, we re-estimated the full model. Since these results are not influenced by outliers as in our previous findings, we see them as more convincing, since OLS models are sensitive to outliers. In our results, we did not find significant relationships with a majority of our variables. However, we found that `AL_tool_Hour`, or the total time spent using AI tools for help with coursework, was significant at the 5% level and negatively correlated with sleep hours. A one-hour increase in AI tool usage led to a 0.0322 decrease in sleep hours. A study on AI-related procrastination in workers found that boredom often led to procrastination via interactions with AI chatbots (Li et al., 2025). We believe that students who use AI tools more often may fall victim to procrastination, thereby increasing the hours spent on AI and potentially cutting into sleep.

The reason it would affect their sleep is based on findings from another study that found schoolchildren were willing to sacrifice sleep hours to complete homework (Sun et al., 2014). This, coupled with increased hours using AI tools and subsequently increased chances of procrastination, would cause students to spend more of their sleeping time completing assignments they postponed.

Additionally, we found that `percent_financial_aid`, or the percentage of financial aid received by students, was also negatively correlated with sleep hours, and significant at the 10% level. A one percentage point increase in financial aid led to a 0.0138-hour decrease in sleep. In one study on undergraduate students experiencing financial stress, researchers found that subjective well-being reports were lower in individuals experiencing significant financial strain (Robb, 2017). Since students receiving financial aid are often under pressure to maintain certain grades, course loads, or participate in activities such as mentorship or sports teams (UBC), this means that high stress levels caused by financial strain might influence a student's requirement for financial aid, and this requirement may then impact their sleep. Stress is known to cause sleep disruptions in a variety of ways, such as through increased nighttime rumination and worsened sleep cycles (Kalmbach et al., 2018). Therefore, financial stress may have a negative impact on student sleeping hours.

11 Variance Inflation Factor (VIF) analysis

Table 9: Variance Inflation Factors (VIF)

Variable	VIF
GPA	1.163
Coursetake.semester	1.096
hour_studentclub	1.091
IsDomestic	1.219
Hour_screen_nonacademic	1.073
Hour_all.courses	1.299
AL_tool_Hour	1.274
hour_volunteer	1.078
percent.financial.aid	1.070
feeling_depress	1.192
Extra_hour_forexam	1.198
hour_sports	1.123
hour_socialize	1.264
Hour_work	1.586
Hour_commute	1.149
year_level	1.070

The overall VIF score of each variable is less than two shows there is multicollinearity among variables.

12 Conclusion

After analyzing our data and selecting the appropriate variables based on multicollinearity and significance tests, we found three variables that had a significant impact on sleeping hours: non-academic screen time, time spent using AI tools for coursework, and the presence of financial aid. For non-academic screen time, we theorized that non-academic screen usage, especially in the evening and before bed, had a negative impact on sleep hygiene. However, this data was not adjusted for outliers, and therefore we were concerned about potentially skewed results.

Then, we trimmed our data and found the other two significant variables, and lost significance for non-academic screen time. We theorized, based on previous research, that AI usage leads to some time being procrastinated interacting with the model, and therefore uses up time students would have otherwise had for homework. Since students value completing coursework, they would have had to sacrifice sleeping hours to complete their procrastinated work.

For financial aid, the stress of being financially insecure as an undergraduate student could have contributed to poor sleeping outcomes, such as through increased time spent ruminating on issues or shortened sleep cycles, which would have decreased hours slept. As our data pool is limited to undergraduate students from a single course, we recommend that further research be conducted on the impacts of these two variables on sleeping hours. Additionally, we believe that our manner of measuring sleep was not ideal: sleep hours are only one aspect of sleep quality, and students may be affected by poor sleep due to the various variables we examined, but in a way our model does not capture.

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